An introduction version control

https://stirlingcodingclub.github.io/version_control

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Using version control

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- Hands-on practice setting up and using version control in your own work with accompanying notes for guidance

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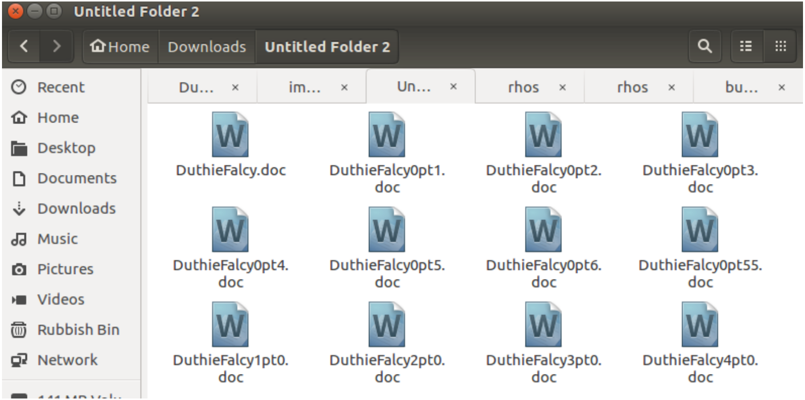
Rough outline of version control

1. What is version control, and why use it?
2. Getting set up – good file management
3. Setting up GitHub
4. The GitHub Desktop tutorial
5. Independent work using version control

What is version control, and why use it?



What version control software does

- Software that records changes you make to files over time
  - Manage different *versions* of files (no need to 'Save As...')
  - Recover old files, keep track of file changes
  - Collaborate with others on shared files

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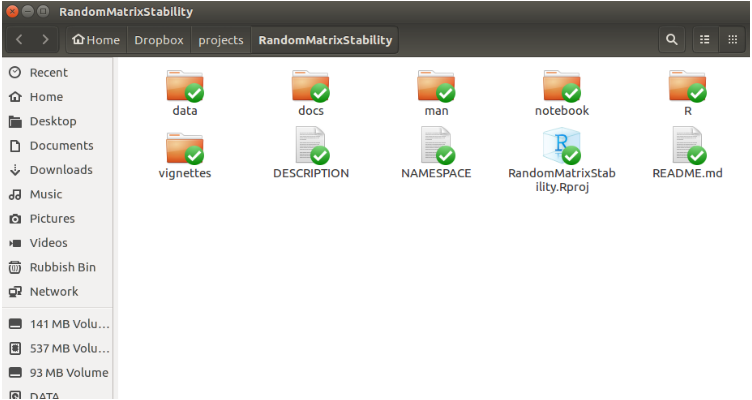
- **Put more intuitively**, version control takes a snapshot in time (called a '**commit**') of all the files in one of your folders (called '**repositories**')
  - Visualise changes to your files over time
  - Look at the differences between file versions
  - Record who changed files, and what they changed

Inside of a project on version control



Folders (a.k.a, 'repositories') include all data files, R code, notes, manuscript drafts, etc.

Full annotated timeline of folder changes (GitHub)

Commits on Mar 1, 2019

Some work on the SI
bradduthie committed on 1 Mar 2019 ✓ — 5d29331

Major restructure and revision of the Discussion to compare to Gibbs … …
bradduthie committed on 1 Mar 2019 ✓ — a40ede5
…et al.

Commits on Feb 27, 2019

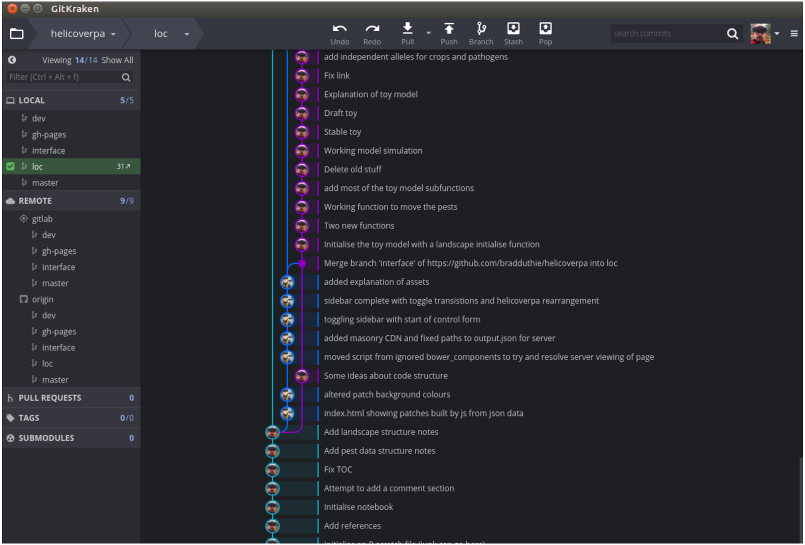
Edit the manuscript up to Reviewer 2 specific comment 3 -- these are … …
bradduthie committed on 27 Feb 2019 ✓ — d411fd5
…next on the list, specifically the new Discussion paragraph

Commits on Feb 22, 2019

First pass, fix some notation
bradduthie committed on 22 Feb 2019 ✓ — 39f854b

Add some to the abstract to emphasise finite systems
bradduthie committed on 22 Feb 2019 ✓ — 612aa4b

Parallel versions (‘branches’) of a folder



Clear breakdown of what has changed

Version control makes science easier

- **Organises files** by avoiding 'save as' multiple versions
  - analysis_1.R
  - analysis_2.R
  - analysis_FINAL.R
  - analysis_FINAL_no_really_this_time.R

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- **Gives peace of mind** to experiment by removing any fear of breaking something that you know works

Version control can help open science

- Transparent record of data collection, analysis, and writing
- Record publicly available on GitHub, Bitbucket, or GitLab
- GitHub repository can be copied, reproduced, and discussed
- git and GitHub can track individual contributions to a project

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- Works across platforms
  - Windows
  - Linux
  - Mac
- Invented by Linus Torvalds